

Syllabus:- Definite Integration

SECTION-I

Straight Objective Type

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- If $I_r = \int_0^{\pi/4} \tan^r x \, dx$ then $I_2 + I_4, I_3 + I_5, I_4 + I_6$ are in

(1) AP (2) GP (3) HP (4) none of these
- $\int_{1/3}^{2/3} \frac{\ln x}{\ln(x-x^2)} \, dx$ is equal to

(1) $1/3$ (2) $1/6$ (3) $1/18$ (4) $1/54$
- If $a_n = \int_0^{\pi} \frac{\sin(2n-1)x}{\sin x} \, dx$ then $a_1, a_2, a_3, \dots, a_n$ are in

(1) AP only (2) GP only
(3) both in AP and GP (4) neither in AP nor in GP
- $\sum_{r=1}^n \int_0^{\pi/2} (r + \sin \theta)^2 \cos \theta \, d\theta =$

(1) $\frac{n}{3} (n^2 + 3n + 3)$ (2) $\frac{n(n+1)(n+2)}{3}$
(3) $\frac{n(n+1)(2n+1)}{3}$ (4) $\frac{(n+1)(n+2)}{3}$
- If $I_1 = \int_0^{\pi/2} \cos(\sin x) \, dx, I_2 = \int_0^{\pi/2} \sin(\cos x) \, dx$ and $I_3 = \int_0^{\pi/2} \cos x \, dx$ then

(1) $I_1 > I_2 > I_3$ (2) $I_3 > I_2 > I_1$ (3) $I_3 > I_1 > I_2$ (4) $I_1 > I_3 > I_2$
- The function $f(x) = \int_1^x [2(t-1)(t-2)^3 + 3(t-1)^2(t-2)^2] \, dt$ attains local maximum at

(1) $x = 0$ (2) $x = 1$ (3) $x = 2$ (4) $x = 4$

7. Let a, b, c be non-zero real numbers such that

$$\int_0^1 (1 + \cos^8 x)(ax^2 + bx + c) dx = \int_0^2 (1 + \cos^8 x)(ax^2 + bx + c) dx$$

then the quadratic equation $ax^2 + bx + c = 0$ has

- (1) no root in (0, 2) (2) atleast one root in (0, 2)
(3) a double root in (0, 2) (4) two imaginary roots

8. The value of integral $\int_0^1 (1 + e^{-x^2}) dx$ is

- (1) -1 (2) 2 (3) $1 + e^{-1}$ (4) none of these

9.
$$\lim_{x \rightarrow \frac{\pi}{4}} \frac{\int_{\frac{\pi}{4}}^{\sec^2 x} f(t) dt}{x^2 - \frac{\pi^2}{16}} =$$

- (1) $\frac{8}{\pi} f(2)$ (2) $\frac{2}{\pi} f(2)$ (3) $\frac{2}{\pi} f\left(\frac{1}{2}\right)$ (4) $4f(2)$

10. The value of $\int_{-\pi/2}^{\pi/2} \frac{\sin^2 x dx}{1 + e^x}$ is-

- (1) $\frac{\pi}{8}$ (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{6}$ (4) $\frac{\pi}{3}$

11. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(2 - x) = f(2 + x)$ and $f(4 - x) = f(4 + x)$ for all $x \in \mathbb{R}$ and $\int_0^2 f(x) dx =$

5. Then the value of $\int_{10}^{50} f(x) dx$ is :

- (1) 125 (2) 80 (3) 100 (4) 200

12. For $x \in \mathbb{R}$, $x \neq 0$, if $y(x)$ is a differentiable function such that $x \int_1^x y(t) dt = (x + 1) \int_1^x ty(t) dt$, then $y(x)$ equals

(where C is a constant)

- (1) $Cx^3 e^{\frac{1}{x}}$ (2) $\frac{C}{x} e^{-\frac{1}{x}}$ (3) $\frac{C}{x^2} e^{-\frac{1}{x}}$ (4) $\frac{C}{x^3} e^{-\frac{1}{x}}$

13. If $f(x) = \int_0^x t(\sin x - \sin t) dt$ then :

- (1) $f'''(x) - f''(x) = \cos x - 2x \sin x$ (2) $f'''(x) + f''(x) - f'(x) = \cos x$
(3) $f'''(x) + f''(x) = \sin x$ (4) $f'''(x) + f'(x) = \cos x - 2x \sin x$

14. The integral $\int_{\sqrt{\ln 5}}^{\sqrt{\ln 7}} \frac{x \cos x^2 dx}{\cos(\ln 35 - x^2) + \cos x^2}$ is equal to
- (1) $\frac{1}{4} \ln \frac{7}{5}$ (2) $\frac{1}{2} \ln \frac{7}{5}$ (3) $\frac{1}{4} \ln \frac{5}{7}$ (4) $\frac{1}{2} \ln \frac{5}{7}$
15. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = \int_0^1 \frac{x^2 + t^2}{2 - t} dt$. Then the curve $y = f(x)$ is -
- (1) a hyperbola (2) an ellipse (3) a straight line (4) a parabola
16. The value of $\int_{-2n}^{2n + \frac{1}{2}} (\sin \pi x) \left\{ \frac{x}{2} \right\} dx$ is (where $\{x\}$ denotes the fractional part of x) -
- (1) $\frac{-2n\pi + 1}{\pi^2}$ (2) $\frac{n}{\pi}$ (3) $\frac{(n+1)}{\pi}$ (4) $\frac{2n\pi - 1}{\pi^2}$
17. The integral $\int_0^{1/2} \frac{e^x (2 - x^2)}{(1 - x)^{3/2} (1 + x)^{1/2}} dx$ is equal to
- (1) $\sqrt{3}e$ (2) $\sqrt{3}e - 1$ (3) $\sqrt{\frac{e}{3}}$ (4) $\sqrt{\frac{e}{3}} - 1$
18. $\int_0^\infty \frac{dx}{(x + \sqrt{1 + x^2})^3} = \alpha$ then find value of 8α .
- (1) 1 (2) 2 (3) 5 (4) 3
19. If $I = \int_0^2 \left(\frac{5 - x}{9 - x^2} \right) dx$ and $J = \int_{10}^{19} \frac{\sin x}{1 + x^8} dx$ then
- (1) $0 < I < 1, J < 10^{-7}$ (2) $1 < I < \frac{6}{5}, J < 10^{-7}$ (3) $I > 2, J > 10^{-7}$ (4) $I < 2, J > 10^{-7}$
20. If $n \in \mathbb{N}$ then $\int_0^\infty x^n e^{-x} dx =$
- (1) $\frac{n!}{2}$ (2) $\frac{n!}{3}$ (3) $n!$ (4) $2(n!)$

SECTION-II

(Integer Type Questions)

This section contains **Five (5)** questions. The answer to each question is **NUMERICAL VALUE** with two digit integer and decimal upto one digit

21. If $\int_0^{\pi} x f(\cos^2 x + \tan^4 x) dx = \frac{\pi k}{1997} \int_0^{\pi/2} f(\cos^2 x + \tan^4 x) dx$ then value of k is -
22. $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$ using this result $\int_0^1 \frac{\ln x}{1+x} dx = -\frac{\pi^2}{k}$. Find value of k/4
23. If $A = \int_0^{\pi} x f(\cos^2 x) dx$ and $B = \int_0^{\pi} f(\cos^2 x) dx$ then $\frac{A}{\pi B} =$
24. Let $\frac{d}{dx} F(x) = \frac{e^{\sin x}}{x}$, $x > 0$. If $\int_1^4 \frac{2e^{\sin x^2}}{x} dx = F(k) - F(1)$ then k may take value
25. If 'a' is a positive integer, then the number of solutions of inequation $\int_0^{\pi/2} \left\{ a^2 \left[\frac{\cos(3x)}{4} + \frac{3}{4} \cos x \right] + a \sin x - 20 \cos x \right\} dx \leq \frac{-a^2}{3}$ is....

Answer Key DPP_16

1.	(3)	2.	(2)	3.	(3)	4.	(1)	5.	(4)	6.	(2)	7.	(2)
8.	(4)	9.	(1)	10.	(2)	11.	(3)	12.	(4)	13.	(4)	14.	(1)
15.	(4)	16.	(1)	17.	(2)	18.	(4)	19.	(2)	20.	(3)	21.	(1997)
22.	(3)	23.	(0.5)	24.	(16)	25.	(4)						